Data Cleaning Log — HeavySuppliersWarehouseDatasets

Corrected 2026-10-28

Corrected version. Changes from the original: (1) §2.1 — current_stock does NOT match the ledger on all 180 rows; (2) §3.1 — duplicate-ID counts corrected to 196 invoice IDs / 201 payment IDs; (3) §3.2 — basis for the P002 correction stated. Every figure below was re-checked against the raw CSVs.

Scope

This log covers every change made to the 12 original CSV files during cleaning, every check run to verify data quality, and every issue found along the way — including which ones were fixed, which were left alone on purpose, and why.

Starting point: 12 files, 605,141 rows, 134 columns. Profiling (see separate profiling report) had already confirmed 0 exact duplicate rows and 0 orphaned foreign keys across 17 tested relationships before any cleaning began.

1. Changes applied to all files

1.1 Date columns converted to proper datetime type

The following columns were being read in as plain text and were converted to actual dates:

- invoices.csv: invoice_date, due_date
- payments.csv: payment_date
- sales_orders_header.csv: order_date, delivery_date
- purchase_orders_header.csv: order_date, expected_delivery_date, received_date
- stock_ledger.csv: movement_date
- customers.csv: customer_since, last_purchase_date
- products.csv: last_purchase_date

1.2 branches.csv — warehouse_capacity split

warehouse_capacity was stored as text with a unit suffix (e.g. “45230 sqft”). The number was extracted into a new column, warehouse_capacity_sqft, and the original text column was dropped.

2. Issues found and left unchanged (documented, not fixed)

2.1 inventory_master.csv — stock threshold scale mismatch [CORRECTED]

current_stock values (roughly 88,853–121,013) are on a completely different scale from max_stock, reorder_level and safety_stock (16–582).

Check against stock_ledger.csv (corrected): the latest running_balance per product/branch equals current_stock on **166 of 180 positions**. The other **14 positions differ by 16 to 515 units** (0.5% or less of the stock level). Differences (current_stock minus ledger balance): P002/CHN001 −297; P004/PUN001 −165; P011/HYD001 +136; P014/AHM001 +299; P018/PUN001 +148; P021/DEL001 −205; P025/HYD001 −515; P026/PUN001 −145; P026/HYD001 −198; P028/DEL001 −223; P028/PUN001 +16; P029/DEL001 −80; P029/PUN001 −58; P029/KOL001 −239.

The original log stated the two matched “exactly on all 180 rows”. That was incorrect. current_stock is close to the ledger but not identical; the ledger balance is the figure used downstream.

The scale mismatch itself stands. Whether the threshold fields are on the wrong scale, or the stock level is overstated (the ledger receives about 18.1× more units than it ships — see Validation Log, ledger vs orders), cannot be settled from the data and is an open question for the source-system owner.

Action taken: No values were changed. Anyone building stockout, reorder, or overstock logic must account for this.

2.2 purchase_orders_header.csv — received_date nulls

received_date is null for all 2,370 rows where po_status = Cancelled, and filled for every one of the 21,630 Received rows — zero exceptions either way. This is expected (a cancelled order was never received), not missing data. Action taken: No values were changed.

3. Issues found and fixed

3.1 invoices.csv / payments.csv — non-unique primary keys [CORRECTED]

invoice_id and payment_id are meant to be unique per row but weren't:

- **196 invoice_id values** are reused across **393 rows** (195 IDs shared by two rows, one ID shared by three). 18,033 rows carry 17,836 distinct IDs, i.e. 197 surplus rows — *the original log reported this surplus (197) as the number of IDs*. Every reused ID belongs to different sales orders (so_id is unique per invoice); 0 exact duplicate rows.
- **201 payment_id values** are reused across **403 rows** (200 shared by two rows, one by three). 19,257 rows carry 19,055 distinct IDs (202 surplus rows). 0 exact duplicate rows.

Confirmed present in the original raw files before any cleaning touched them — a source data defect, not something introduced during cleaning.

Fix applied: Added a new, guaranteed-unique surrogate key to each file — invoice_uid and payment_uid (row-based IDs, e.g. INVUID-000001). These do not derive from the original ID columns, so they cannot collide. The original invoice_id / payment_id columns were left completely untouched for reference.

Downstream consequence found and flagged, not fixed: because payments.csv links to invoices only via the (non-unique) invoice_id, 417 payment rows and 393 invoice rows cannot be confidently matched to a single invoice. Rather than guess, a new boolean column, invoice_id_ambiguous, was added to both files, True on every affected row. This surfaced as two symptoms during validation before the root cause was traced:

- 247 Paid invoices where summed payments didn't match grand_total
- 35 Unpaid invoices that had a payment record attached

Both symptoms are fully explained by the ID collision (100% overlap with the flagged rows) — they are not separate defects.

3.2 products.csv — one incorrect margin_percentage [CLARIFIED]

P002 (Engine Oil Filter OF-90) listed margin_percentage = 37.3, but its unit_cost (450) and unit_price (720) give **60.0**. **Corrected to 60.0** (confirmed in both products_clean.csv and the Merged datasets products.csv).

Why 60.0 is the right value: in this dataset margin_percentage is a **markup on cost**, (price − cost) / cost × 100. That formula reproduces the stored value for 29 of the 30 products to within ±0.1 (P001, P005 and P014 differ by 0.07–0.10 — rounding, left unchanged). P002 is the only outlier: 37.3 is close to the price-basis gross margin (720 − 450) / 720 = 37.5%, i.e. it appears to have been calculated on a different basis from the other 29 products.

Naming caution: because the column is a markup, it must not be used as "Gross Margin %". Gross margin on price is (price − cost) / price (P002: 37.5%).

4. Validation checks run (no issues found)

- Primary/composite key uniqueness checked on all 12 tables: branches, customers, suppliers, products, purchase_orders_header, sales_orders_header, stock_ledger (single-column keys), plus inventory_master (product_id+branch_id), purchase_orders_lines (po_id+line_number), sales_orders_lines (so_id+line_number) — all clean except invoices/payments (see 3.1).
- customer_rating and reliability_score within expected 1–5 range — no violations.
- No customer over their credit_limit.
- No negative or zero values in quantity, cost, price, or payment fields.
- No backwards date ranges (order date after delivery date, etc.).
- Header-level tax math (total + GST = grand_total) consistent across invoices, sales_orders_header, and purchase_orders_header — no mismatches.

5. Files affected by this cleaning pass

File	Changed?	What changed
branches.csv	Yes	warehouse_capacity split into warehouse_capacity_sqft
customers.csv	Yes	Date columns converted
suppliers.csv	No	—
products.csv	Yes	Date column converted; P002 margin_percentage 37.3 → 60.0
inventory_master.csv	No	Scale mismatch and 14-position ledger variance documented, not altered
purchase_orders_header.csv	Yes	Date columns converted
purchase_orders_lines.csv	No	—
sales_orders_header.csv	Yes	Date columns converted
sales_orders_lines.csv	No	—
invoices.csv	Yes	Date columns converted; invoice_uid and invoice_id_ambiguous added
payments.csv	Yes	Date column converted; payment_uid and invoice_id_ambiguous added
stock_ledger.csv	Yes	Date column converted